

MID TERM EXAMINATION

Course: Data Structures (CS101)

Set: B | Time: 3 Hours | Max Marks: 40 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
ALL

SECTION A: MCQs

1. Which SQL keyword is used to retrieve data from a database?

[2 Marks]

- a) UPDATE
- b) INSERT
- c) SELECT
- d) DELETE

2. A database constraint that ensures that a value in one table's column(s) must match a value in another (parent) table's primary key column(s) is known as a:

[2 Marks]

- a) Primary key constraint
- b) Unique constraint
- c) Foreign key constraint
- d) Not Null constraint

3. Which of the following is an example of a DDL (Data Definition Language) command?

[2 Marks]

- a) INSERT
- b) UPDATE
- c) DELETE
- d) ALTER

4. The phenomenon where a transaction reads uncommitted data written by another transaction, which then subsequently rolls back, is known as:

[2 Marks]

- a) Lost Update
- b) Dirty Read
- c) Non-repeatable Read
- d) Phantom Read

5. In the context of a relational database, what does a "tuple" refer to?

[2 Marks]

- a) A column in a table
- b) A row in a table
- c) A database schema
- d) An index

6. Which of the following database components is responsible for managing the allocation of space on disk storage and the data structures used for data storage?

[2 Marks]

- a) Query Processor
- b) Transaction Manager
- c) Storage Manager
- d) Buffer Manager

SECTION B: SHORT ANSWERS

7. Explain the role of a Database Administrator (DBA) in managing a DBMS. List two key responsibilities. [6 Marks]

8. Differentiate between a DDL (Data Definition Language) and a DML (Data Manipulation Language) in a DBMS, providing an example command for each. [5 Marks]

9. What is a database schema? Briefly explain the three-level architecture (ANSI/SPARC architecture) for a DBMS. [4 Marks]

10. Describe the "Lost Update" problem that can occur in a multi-user database environment without proper concurrency control. Provide a simple example scenario. [7 Marks]

11. Briefly explain the purpose of database normalization. Name and briefly describe the first normal form (1NF). [6 Marks]

--- End of Question Paper ---